THE Open GROUP

TOGAF® Standard – Architecture Roadmap

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Barangay San Antonio Health Department Enterprise Architecture

Barangay San Antonio Health Center, Pasig City

Document Version History

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Tracking Information

Project Name	Barangay San Antonio Health Department Enterprise Architecture		
Prepared By	Rome Victor Mag-iba Sendin (Group Leader); Samuel Anthonius Badillo; Wilson Halim; Kevin Enrico Ang	**Document Version No**	1.0
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Distribution List

From	Date	Phone/Email
Rome Victor Mag-iba Sendin, Group Leader, UA&P Solution Architecture	October 22, 2026	romevictor.sendin@uap.asia / 09456315746

To	Action*	Due Date	Phone//Email
Dr. Glynnis Lynn U. Bonilla, Medical Officer IV	Approve	October 23, 2026	bsahealthcenter@gmail.com
Hon. Thomas Raymond U. Lising, Punong Barangay	Review	October 23, 2026	bsahealthcenter@gmail.com
Hon. Rachel Marie S. Rustia-Olarte, Co-Chair, Committee on Health & Nutrition	Review	October 23, 2026	bsahealthcenter@gmail.com
Mr. Joel Pira, Instructor, Solution Architecture (UA&P)	Review	October 23, 2026	joel.pira@uap.asia

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1. Purpose of Document

The Architecture Roadmap lists individual work packages that will realize the Target Architecture and lays them out on a timeline to show progression from the Baseline Architecture to the Target Architecture. The Architecture Roadmap highlights individual work packages' business value at each stage. Transition Architectures necessary to effectively realize the Target Architecture are identified as intermediate steps. The Architecture Roadmap is incrementally developed throughout Phases E and F, and informed by readily identifiable roadmap components from Phase B, C, and D within the ADM.

1.1 Output and Input

The TOGAF® Content Framework (see Architecture Deliverables) identifies deliverables that are produced as outputs from executing the Architecture Development Method (ADM) cycle and potentially consumed as inputs at other points in the ADM.

This deliverable is produced, updated, and consumed by the ADM Phases:

Deliverable	Output from…	Input to…
Architecture Roadmap	B, C, D, E, F	B, C, D, E, F

Tailoring note: the table above shows the Architecture Roadmap is produced across Phases B to F. In this engagement it was started in Phase B (candidate components from the business gaps), refined in Phases C and D, and completed in Phase E, so the final version is filed with the Phase E deliverables.

2. Candidate Roadmap Components

Candidate components are based on the gaps between the Baseline and Target Architectures found in Phases B, C and D.

2.1 Business Architecture Candidate Roadmap Components

- Medicine buffer policy and stock monitoring (WP1)
- Records filing standard and custodian procedures (WP4)
- Single-tally reporting routine (WP5)
- Patient queue process (WP6)
- Condo and health-promotion partnerships (WP9)

2.2 Information Systems (Data and Application) Architecture Roadmap Components

- Shared stock tracker (WP1)
- Official workspace and shared storage (WP2)
- Active-chart index (WP4)
- Report templates from one tally source (WP5)
- Queue application (WP6)
- Requested EMR–YAKAP and dental interoperability (WP8)

2.3 Technology Architecture Roadmap Components

- Shared computers for aides and BHWs (WP3)
- Backup internet with failover (WP7)
- Queue ticket and display hardware (WP6)
- BNS field laptop and weighing kit (WP9)
- City laptop replacement (WP8)

3. Initial Complete Version of the Architecture Roadmap

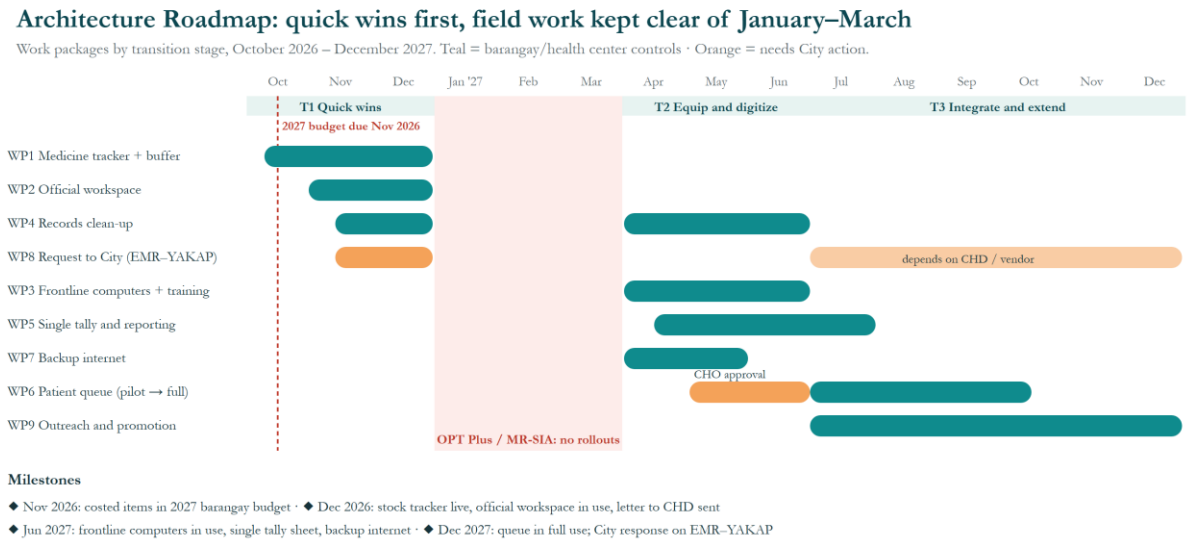


Figure 1. Architecture Roadmap, October 2026 – December 2027.

How to read it: time runs left to right from October 2026 to December 2027. Each bar is a work package; teal = barangay/health center controls, orange = needs City action. The shaded band is the January–March no-rollout window; diamonds are milestones; T1–T3 are the transition stages.

- **T1 Quick wins (Nov–Dec 2026):** stock tracker and buffer (WP1), official workspace (WP2), records clean-up starts (WP4), letter to the City (WP8); costed items go into the 2027 budget before the November deadline.
- **Freeze (Jan–Mar 2027):** no changes during OPT Plus and MR-SIA.
- **T2 Equip and digitize (Apr–Jun 2027):** frontline computers and training (WP3), single tally sheet (WP5), backup internet (WP7), queue pilot after CHO approval (WP6).
- **T3 Integrate and extend (Jul–Dec 2027):** queue in full use, outreach expanded (WP9), City response on EMR–YAKAP integration.
- **Takeaway:** the plan does not wait on the City; orange items run in parallel.

3.1 Work Package Portfolio

Nine work packages realize the Target Architecture.

3.1.1 Work Package Description (Name, Description, Objectives, Deliverables)

ID	Name	Description, objectives and deliverables	Owner
WP1	Medicine stock tracker and buffer stock	Shared stock sheet (beginning balance, received, issued, ending balance, reorder level) updated daily; standing barangay buffer line for maintenance medicines (Metformin, Losartan, Amlodipine ran short in 2026)	Nurse Hardman; Kagawad Rustia

ID	Name	Description, objectives and deliverables	Owner
WP2	Official workspace and communication	Barangay-owned Google account or Workspace with a shared drive for program files; work-only chat groups; storage for OPT files	Barangay IT; Kagawad Rustia
WP3	Frontline computers and digital skills	2–3 shared desktops for the 4 Pasig Health Aides and 3 BHWs; basic computer workshops (UA&P outreach)	Kapitan Lising; UA&P team
WP4	Records clean-up and security	Active-chart index by record number for the ~8,000 charts; archive inactive charts on the DOH 7-year schedule; lockable records cabinet and key log; request consent once per patient (CHD approval)	Ms. Roa (custodian); Ms. Abong
WP5	Single tally and reporting	Aides and BHWs enter daily tallies once on the shared computer; program reports built from that sheet; request CHD to accept digitally signed reports	Ms. Abong; Nurse Hardman
WP6	Patient queue	Numbered tickets and display for consultation days; built as a UA&P IT capstone pilot after City Health Officer approval	Dr. Bonilla; UA&P IT program
WP7	Backup connectivity	Second provider or LTE router with failover alongside the PLDT line; written outage procedure (paper fallback)	Barangay IT
WP8	City interoperability request	Evidence-based letter to CHD: EMR–PhilHealth YAKAP prescription link, dental eMED link, replacement of 10-year-old City laptops, digital report submission	Dr. Bonilla; Kapitan Lising
WP9	Outreach and health promotion	Condo partnerships through the Community Wellness Pop-ups; oral health campaigns; BNS field laptop and complete weighing kit	Kagawad Rustia; Dr. Magpali; Mr. Vasquez

3.1.2 Functional Requirements

WP	Requirements addressed
WP1	R-02 stock tracking with alerts; R-03 buffer budget
WP2	R-05 official workspace
WP3	R-04 computers for aides/BHWs; R-11 training
WP4	R-09 paper fallback; R-10 privacy and retention
WP5	R-07 fewer redundant forms
WP6	R-06 patient queue
WP7	R-08 backup internet
WP8	R-01 single prescription entry
WP9	R-12 health promotion

3.1.3 Dependencies

- WP5 depends on WP2 (shared storage) and WP3 (computers).
- WP6 depends on City Health Officer approval and WP7 (stable internet).
- WP8 outcomes depend on the City, its EMR vendor and PhilHealth.
- WP1 buffer and WP3, WP7, WP9 purchases depend on inclusion in the 2027 barangay budget (proposals due November 2026).

3.1.4 Relationship to Opportunity

Each work package addresses one of the Phase A goals: enter data once (WP5, WP8), keep medicines in stock (WP1), faster records and reports (WP4, WP5), official secure tools (WP2, WP3, WP7), and better patient flow and reach (WP6, WP9).

3.1.5 Relationship to Architecture Definition Document and Architecture Requirements Specification

WP	Gaps closed	Success measure (ARS)
WP1	BG-2, BG-3, DG-5, AG-3	SM-2
WP2	DG-4, AG-5, TG-4	SM-6
WP3	BG-7, TG-3	SM-7
WP4	BG-4, DG-2, DG-7, DG-8, TG-7	SM-3
WP5	BG-5, DG-3, AG-6	SM-4, SM-5
WP6	BG-6, AG-4, TG-5	Queue complaints
WP7	TG-1	Outages handled without paper-only days
WP8	BG-1, DG-1, AG-1, AG-2, TG-2	SM-1
WP9	BG-8, TG-6	Program coverage

3.1.6 Business Value

WP	Value
WP1	Patients keep receiving maintenance medicines; barangay spending becomes planned
WP2	Work data protected and shared; no staff paying for storage
WP3	Frontline staff record their own work; less load on nurse, midwife and encoder
WP4	Faster registration; legal compliance; space freed from inactive charts
WP5	Reports finished in working hours; fewer errors
WP6	Fairer, calmer waiting; fewer complaints
WP7	Consultations continue through outages
WP8	Up to two fewer entries per prescription if adopted City-wide
WP9	Higher reach of condo residents and preventive care

3.2 Implementation Factor Catalog

Factors affecting implementation are listed below.

3.2.1 Risks

Risk	Mitigation
2027 budget proposals (due November 2026) submitted without the costed items	Brief Kapitan and Kagawad with the cost table
City declines or delays EMR–YAKAP integration	Proceed with barangay-controlled packages; keep requesting with evidence
Low digital skills of aides and BHWs	Hands-on workshops; simple sheet design
Rollout clashes with OPT Plus (Jan–Mar) or month-end reporting	Schedule per roadmap
Medicine shortage recurs before the buffer is funded	Interim purchase via Kagawad, tracked in WP1

3.2.2 Issues

- Cross-barangay visibility of EMR records is unconfirmed (to ask Mr. Reyes).
- Status of the City laptop replacement request is unknown.
- Governance view differs: barangay autonomy (Kapitan) vs. City approval for systems (Dr. Bonilla).

3.2.3 Assumptions

- Staffing stays the same in 2027.
- The City EMR remains the system of record.
- Cost figures are indicative.

3.2.4 Dependencies

See Section 3.1.3 (work package dependencies). External dependencies: City Health Department, EMR vendor, PhilHealth, barangay budget approval.

3.2.5 Actions

- Present findings and roadmap to Dr. Bonilla (October 23, afternoon).
- Give Kapitan and Kagawad the cost table before the November 2026 budget deadline.
- Send the WP8 letter to CHD through Nurse Hardman.
- Inform the IT Program Director of the WP6 capstone opportunity.

3.2.6 Inputs

Interview Findings (file 11); Architecture Vision; Architecture Definition Document; Architecture Deliverables for Phases B, C and D; Architecture Requirements Specification.

3.3 Consolidated Gaps, Solutions, and Dependencies Matrix

Architecture domain	Gap	Potential solutions	Dependencies
Business	BG-1 to BG-9: triple entry, manual stock, ad-hoc buying, mixed charts, after-hours and hand-carried reports, no queue, no frontline devices, condo resistance, clinician paperwork	WP1, WP3, WP4, WP5, WP6, WP9	Barangay budget; CHO approval
Data	DG-1 to DG-8: triple entry, inactive charts, paper tallies, personal storage, manual stock, no referral log, verbal access, consent every visit	WP1, WP2, WP4, WP5, WP8	WP2 storage; CHD approval for consent
Application	AG-1 to AG-7: EMR–YAKAP and dental links, stock tool, queue, workspace, retyped tallies, booking	WP1, WP2, WP5, WP6, WP8	City / vendor / PhilHealth
Technology	TG-1 to TG-7: single line, old laptops, no frontline computers, no official storage, no queue hardware, no field laptop, partial lock	WP2, WP3, WP4, WP6, WP7, WP8, WP9	Barangay budget; City for laptops

3.3.1 Architecture Domain

Business, Data, Application and Technology (see the matrix above).

3.3.2 Gap

Gap IDs BG, DG, AG and TG are defined in the Architecture Deliverables for Phases B, C and D.

3.3.3 Potential Solutions

Work packages WP1–WP9 (Section 3.1.1) and building blocks (E2, E3).

3.3.4 Dependencies

See the matrix above and Section 3.1.3.

3.4 Transition Architectures

See the Architecture Definition Document, Section 7.

State	Period	Description
Baseline	October 2026	As described in Phases B–D
T1 Quick wins	November–December 2026	Stock tracker, official workspace, chart index started, City request letter sent, costed items proposed in the 2027 barangay budget (due November 2026)
Freeze	January–March 2027	No changes during OPT Plus and MR-SIA
T2 Equip and digitize	April–June 2027	Shared computers, single tally sheet, backup internet, queue pilot, inactive charts archived
T3 Integrate and extend (target)	July–December 2027	Queue in full use, expanded condo outreach, City decision on EMR–YAKAP integration

3.5 Implementation Recommendations

Proceed with barangay-controlled quick wins first, keep January–March free of changes, and pursue City-level changes in parallel.

3.5.1 Criteria Measures of Effectiveness of Projects

- Measured against success measures SM-1 to SM-8 in the Architecture Requirements Specification.
- Reviewed quarterly by the proposed change board (Dr. Bonilla, Kagawad Rustia, Nurse Hardman).

3.5.2 Risks and Issues

See the Implementation Factor Catalog (Section 3.2).

3.5.3 Solution Building Blocks

See E3 – Solution Building Blocks, which maps each SBB to its Architecture Building Block (E2).

A Abbreviations

Abbreviations, acronyms and reference codes used in this document and its figures, in alphabetical order.

Abbreviation	Meaning
ADM	Architecture Development Method, the step-by-step method of the TOGAF Standard
AG-1 …	Application gap reference number (Phase C)
ARS	Architecture Requirements Specification
BG-1 …	Business gap reference number (Phase B)
BHW	Barangay Health Worker
BNS	Barangay Nutrition Scholar
CHD	City Health Department (Pasig City)
CHO	City Health Officer
DG-1 …	Data gap reference number (Phase C)
DOH	Department of Health
eMed / eMED	Name of the City's electronic medical record system; “dental eMED” is its separate dental system
EMR	Electronic Medical Record
Hon.	Honorable (title of elected officials)
IT	Information technology
LTE	Long-Term Evolution (4G mobile internet)
MR-SIA	Measles-Rubella Supplemental Immunization Activity
OPT / OPT Plus	Operation Timbang Plus: yearly weighing and measuring of children aged 0–59 months
PhilHealth	Philippine Health Insurance Corporation
PLDT	PLDT Inc., an internet and telephone provider
R-01 …	Stakeholder requirement reference number
SBB	Solution Building Block (SBB-1 to SBB-9 are reference numbers)
SM-1 …	Success measure reference number
T1 … T3	Transition architecture stage
TG-1 …	Technology gap reference number (Phase D)
TOGAF	The Open Group Architecture Framework (the TOGAF® Standard)
UA&P	University of Asia and the Pacific
WP1 … WP9	Work package (a group of related changes delivered together)
YAKAP	PhilHealth Yaman ng Kalusugan Program (primary-care benefit)